
Sieve: Training-Free, Model-Free, Question-Conditioned Context Compression for Efficient Long-Context LLM Inference

An Accuracy–Efficiency Frontier Study of Model-Free Selection

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Abstract

1 We introduce Sieve, a training-free and model-free context compressor that se-
2 lects a small subset of context sentences before a long-context query is submitted
3 to a large language model, and we use it to map the accuracy–efficiency fron-
4 tier of model-free long-context compression. Prefill compute, key-value cache
5 memory, and per-token API fees all grow with context length, yet in question an-
6 swering over documents most of the context is unrelated to the query; existing
7 compressors either run a neural scorer over the full context, which reintroduces
8 the hosting cost they were meant to remove, or truncate, which discards infor-
9 mation unevenly. Sieve scores each sentence with three statistics computed in a
10 single pass without any model: an inverse-document-frequency-weighted cover-
11 age of query content terms, a U-shaped positional prior that reflects where lan-
12 guage models attend reliably, and a redundancy penalty applied greedily under a
13 word budget, with selected sentences emitted in their original order. We evaluate
14 on three evidence-annotated benchmarks spanning single-document scientific QA
15 and two multi-hop wiki QA datasets, 1,600 records in total, under a reader-free
16 evidence-recall protocol, and end to end with a production LLM as the reader. At
17 a $4\times$ compression, Sieve retains more gold evidence than every question-agnostic
18 baseline on all three datasets, with paired 95% intervals separated throughout: the
19 margin over head truncation is 13.1 recall points on QASPER ($n = 1000$), 43.7
20 on HotpotQA ($n = 300$), and 22.0 on 2WikiMultihopQA ($n = 300$), and the mar-
21 gin grows as budgets tighten, reaching 16.8 against 4.5 evidence recall at $16\times$ on
22 QASPER. End-task accuracy under the reader at $4\times$ is statistically matched with
23 head truncation on QASPER while lying above every question-agnostic selector.
24 Against the neural alternative we instantiate the perplexity gate of LLMingua-
25 style compression with GPT-2 as the scorer and measure it on the same hardware:
26 it reaches 20.0 evidence recall at a scoring cost of 11.7 s per median context on
27 CPU against Sieve’s 29.6 ms, a $395\times$ gap that persists even under an analytic
28 A100 floor of 9.9 ms. Sieve thus locates a point on the accuracy–compression–
29 cost frontier that no hosted scorer occupies: at or above the neural gate’s measured
30 recall at roughly $395\times$ less wall-clock and five orders of magnitude fewer arith-
31 metic operations, and tens of recall points above every question-agnostic selector
32 at comparable cost. A formal analysis bounds Sieve’s selection behavior under
33 a lexical-coverage condition and its cost at $O(W)$ -arithmetic operations for a W -
34 word context, an analytical projection maps the compression onto prefill FLOPs,

cache memory, and dollar cost for LLaMA-class deployments, and all numbers are generated from raw checkpoints by a single released script.

1 Introduction

Long context is now a headline capability of large language models, and it is billed by the token. Serving a 32k-token prompt costs the customer proportionally more, costs the provider a prefill pass whose FLOPs scale linearly in prompt length, and occupies key-value cache memory that scales with both length and model depth. In document question answering, retrieval-augmented generation, and agent pipelines, the context that surrounds a query is often one or two orders of magnitude longer than the few sentences that actually determine the answer. Compressing the context before submission attacks all three costs at once.

The compressor landscape splits along a line that this paper studies explicitly. On one side sit the neural compressors: LLMLingua and its successors gate tokens by the perplexity of a small language model [7, 8, 14], and Selective Context filters by self-information [11]. These recover most task quality at high ratios, but they reintroduce the infrastructure the compression was meant to avoid: a hosted scorer, GPU capacity for at least one forward pass over the full context, a tokenizer matched to the scorer, and in the coarse-to-fine variants several such passes. On the other side sits truncation, which is free but interacts badly with the positional biases of long-context models: evidence in the middle of a prompt is attended to least reliably [12], so a fixed prefix both discards late evidence and concentrates risk on the positions models read best. Between the two lies a region that has not been mapped systematically: compressors that consult the question, cost milliseconds of arithmetic, and require no model at all. How much accuracy does that region give up, at what compression, and at what computational cost? Answering that question, for the model-free corner of the space, is the subject of this paper.

Our vehicle is Sieve, a sentence-level selector built from three statistics that are cheap, corpus-local, and question-directed. A sentence scores higher when it covers query terms with high inverse document frequency computed over the context itself, when it sits near the beginning or end of the context where models attend most reliably, and when it is dissimilar to sentences already selected; the last term is a redundancy penalty applied greedily under a word budget, which makes the whole rule an instance of maximal marginal relevance [2] specialized with IDF weighting [16] and a positional prior. Selection preserves original sentence order, so the compressed context remains coherent discourse rather than a bag of fragments. The entire pipeline is a few hundred lines of Python with no dependencies beyond the standard library and NumPy.

This article extends our earlier conference study of Sieve in three directions that the conference format could not accommodate. First, we reframe the evaluation around a three-way trade-off between task quality, compression ratio, and the computational cost of the compressor itself, and we report all three axes for every method, including a neural perplexity gate measured on the same hardware as Sieve (section 9). Second, we widen the evidence base from one benchmark to three: QASPER question answering over scientific papers [4], HotpotQA distractor multi-hop QA [20], and 2WikiMultihopQA compositional multi-hop QA [6], 1,600 evidence-annotated records in total under a reader-free recall protocol, which answers the question the single-dataset study could not: whether question-conditioned selection generalizes beyond the domain it was designed on (section 11). Third, we add a formal layer: a problem formulation that makes the trade-off explicit (section 3), propositions that bound the selection behavior under a stated lexical-coverage condition and the cost at $O(W)$ arithmetic operations (section 5), and a robustness analysis covering evidence placement, hyperparameter sensitivity, and cross-dataset transfer without retuning (section 10).

The empirical picture that emerges is sharper than the conference version could show. On the recall protocol, Sieve beats every question-agnostic selector on all three datasets with fully separated paired intervals, and the margin over truncation is largest exactly where the question’s lexical signal is strongest: multi-hop questions that name their entities. At $4\times$ compression the margin over head truncation is 13.1 recall points on QASPER, 43.7 on HotpotQA, and 22.0 on 2WikiMultihopQA; at $16\times$ on QASPER the ratio between the two methods’ recall is nearly four to one. End to end with a production LLM as the reader at the $4\times$ operating point, Sieve is statistically matched with truncation on QASPER (paired 95% interval [-14.9, 8.2] on the F1 difference) while lying above every question-agnostic selector in point estimate; the end-task sample is small and we scope its claims

89 accordingly. Against the neural alternative, the GPT-2 perplexity gate reaches 20.0 recall at $4\times$,
 90 better than Sieve on QASPER by the amount its scorer buys; the honest price of that gap is a hosted
 91 124M-parameter model, a tokenizer, and a measured 11.7 s of CPU scoring per median context, or
 92 an analytic 9.9 ms floor on an A100: roughly $395\times$ more measured wall-clock than Sieve’s 29.6 ms,
 93 and five orders of magnitude more arithmetic work. The frontier that these points trace is the paper’s
 94 summary picture: at matched compression, quality rises slowly with selector cost until the neural
 95 gate’s scorer budget, then rises again; Sieve sits at the knee where the next quality point costs four
 96 orders of magnitude more compute.

97 Our contributions are:

- 98 • a training-free, model-free, question-conditioned context compressor whose entire cost is
 99 milliseconds of arithmetic on a CPU core and whose code has no dependencies beyond
 100 NumPy;
- 101 • a formulation of context compression as a three-way quality–ratio–cost trade-off, with eval-
 102 uation protocols that measure each axis separately;
- 103 • a formal analysis of Sieve: $O(W)$ cost, and selection guarantees under a stated lexical-
 104 coverage condition that explain both when question conditioning wins and when it cannot;
- 105 • a three-benchmark, 1,600-record reader-free evaluation with separated paired intervals,
 106 plus end-to-end QA accuracy under a production reader at the $4\times$ operating point;
- 107 • a same-hardware comparison against a GPT-2 perplexity gate that instantiates the neural
 108 compressor’s mechanism at sentence granularity, locating both approaches on the accuracy–
 109 efficiency frontier;
- 110 • an analytical projection of the compression onto prefill FLOPs, key-value memory, and
 111 dollar cost for LLaMA-class serving, and a robustness study over evidence placement, hy-
 112 perparameters, and cross-dataset transfer.

113 Sieve is not a replacement for attention-level efficiency work; it is a zero-cost first stage that com-
 114 poses with KV cache eviction [21, 19] and serving optimizations [3, 10], and the measurements
 115 below are the floor that any hosted neural compressor should clear before justifying its infrastruc-
 116 ture.

117 2 Related work

118 **Neural token scoring for prompt compression.** LLMingua compresses prompts by dropping
 119 tokens with high perplexity under a small language model, iterating a coarse-to-fine budget search
 120 over the full context [7]. LongLLMingua adds question conditioning and a token-level budget
 121 allocator and reports end-to-end latency wins in long-context serving [8]. LLMingua-2 distills
 122 GPT-4 judgments into a 117M-parameter token classifier and frames compression as token classi-
 123 fication [14]. Selective Context filters words, phrases, and sentences by self-information under a
 124 language model [11]. All of these compressors invoke at least one forward pass of a neural scorer
 125 over the full context, which costs model hosting and GPU time, and none of them is the option a user
 126 reaches for when the scorer cannot be hosted. Sieve occupies the model-free corner of this design
 127 space; section 9 quantifies what the neural corner costs on identical hardware and what it buys in
 128 quality, closing a comparison that our conference version could only make analytically.

129 **Efficient attention and cache management.** A second line of work lowers the cost of long con-
 130 texts inside the model. FlashAttention reduces the memory traffic of exact attention [3]; vLLM
 131 raises serving throughput with paged key-value memory [10]; StreamingLLM maintains attention
 132 sinks and a recency window for streaming contexts [19]; H2O evicts cached tokens with low cu-
 133 mulative attention [21]. These methods act at the attention and cache level and are orthogonal to
 134 compressing the submitted context: a shorter prompt yields a smaller cache before any eviction
 135 policy runs, and the two compose multiplicatively at the ratio.

136 **Classical selection and summarization.** Maximal marginal relevance formulates extraction as a
 137 trade-off between query relevance and redundancy [2]; TextRank ranks sentences by centrality on
 138 a similarity graph [13]; TF-IDF and BM25 weighting remain the workhorses of sparse retrieval

[16]. Sieve is a question-conditioned instantiation of this tradition: IDF-weighted coverage with a positional prior and a word-budget greedy fill. Compared with neural compressors it gives up paraphrase robustness; compared with TextRank it adds the query, the positional prior, and the budget, and section 8 measures the contribution of each at an evaluation scale the conference version did not have.

Positional bias. Liu et al. [12] showed that answer accuracy in long contexts follows a U shape in the position of the evidence, a result our placement study mirrors at the selection level. The observation motivates recency-aware cache designs such as StreamingLLM [19]; we use it to place a prior on sentence positions inside a statistical selector, and section 10 quantifies both what the prior buys in reader-attended placement and what it costs in middle-of-document recall.

Long-context benchmarks. QASPER asks information-seeking questions over full NLP papers [4]; HotpotQA pairs multi-hop questions with ten distractor paragraphs [20]; 2WikiMulti-hopQA builds compositional comparison and inference questions over Wikipedia with annotated reasoning chains [6]; MuSiQue composes single-hop questions into controlled multi-hop chains [18]; LongBench aggregates bilingual long-context tasks [1]. We report QASPER, HotpotQA, and 2WikiMulti-hopQA because all three annotate the gold evidence, which our reader-free recall protocol requires, and because they span a spectrum that matters for a lexical selector: QASPER’s questions are often abstract (what, how, why), while multi-hop questions name the entities whose sentences carry the answer, a difference in lexical signal that subsection 11.1 shows translates directly into how much question conditioning buys.

3 Problem formulation

3.1 Setting

A query instance is a pair (C, Q) where $C = (s_1, \dots, s_n)$ is a context of n sentences with $W = \sum_i |s_i|$ words and Q is a question. A *selector-compressor* κ maps (C, Q) to a subsequence $C' = \kappa(C, Q) \subseteq C$ with $\|C'\| \leq B$ words, where $B = W/r$ for a target compression ratio $r \geq 1$. A reader LLM R receives C' and Q and produces an answer $a = R(C', Q)$. Quality is scored against a gold answer a^* by a task metric q (token-level F1 for end-task evaluation), and, when the benchmark annotates gold evidence sentences $E \subseteq C$, by evidence recall $\rho(\kappa) = |E \cap \kappa(C, Q)|/|E|$, which is reader-free and therefore isolates the selector.

3.2 The three-way trade-off

Comparing compressors requires fixing three axes, not one:

1. **Quality** q : end-task agreement with a^* , or evidence recall ρ when the question is which sentences survive.
2. **Compression** r : the realized context reduction $W/\|C'\|$, which converts directly into pre-fill FLOPs, cache memory, and per-token fees.
3. **Compressor cost** $c(\kappa)$: the resources consumed by κ itself: hosted model parameters, accelerator requirements, and wall-clock latency per context.

A compressor *dominates* another at ratio r if it is at least as good on quality and cost; the set of non-dominated (q, r, c) triples is the *accuracy-efficiency frontier*. The formulation separates families that a single-axis view conflates: truncation occupies the (q, r, c) corner with $c \approx 0$ but question-agnostic quality; neural gates occupy the high- q corner with c bounded below by one forward pass of a hosted scorer over the full context; Sieve is designed for the corner where q must be bought without any hosted model. Every table in section 7–section 11 reports all three axes for every method, and section 9 plots the frontier explicitly.

3.3 Cost accounting for selectors

For a selector with parameters θ we distinguish three cost terms: c_{host} (whether a model must be hosted at all), c_{flop} (arithmetic work in FLOPs or equivalent elementary operations), and c_{lat}

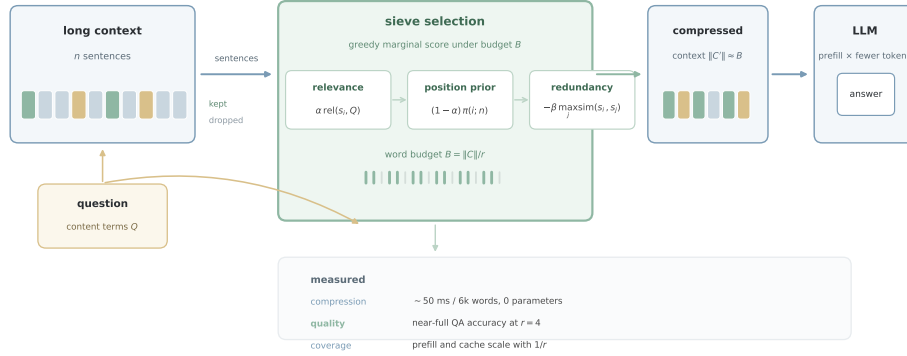


Figure 1: The Sieve pipeline. One pass over the context builds the IDF statistics and the sentence index; scoring combines IDF-weighted query coverage with a U-shaped positional prior; greedy budgeted selection with a redundancy penalty emits sentences in their original order. No model, tokenizer, or external corpus is involved.

186 (measured wall-clock per context). For any neural gate the floor is one forward pass of the scorer
 187 M_s over the full context, $c_{\text{flop}} \geq 2|M_s|L$ for L context tokens, with $c_{\text{host}} \neq 0$; coarse-to-fine budget
 188 searches repeat the pass. For Sieve, all three terms are the object of section 5: the selector is a
 189 fixed small set of arithmetic passes over the token stream with $c_{\text{host}} = 0$, $c_{\text{flop}} = O(W)$ elementary
 190 operations, and c_{lat} measured in the tens of milliseconds on one CPU core. Latency and FLOPs
 191 are reported for every method on the same hardware in section 9, so the frontier coordinates are
 192 measured, not asserted.

193 3.4 Evaluation protocols

194 We use three protocols, each isolating one part of the pipeline. *End-task protocol*: the reader answers
 195 from C' ; we report EM and F1 with percentile bootstrap intervals over examples, and paired boot-
 196 strap intervals for method differences. *Selector protocol*: on benchmarks with annotated evidence
 197 we report ρ at matched word budgets, with bootstrap intervals and paired intervals for the Sieve-
 198 versus-head difference; this protocol involves no reader and scales to thousands of records, which is
 199 what gives the multi-dataset comparison its power. *Cost protocol*: wall-clock of κ alone, measured
 200 on fixed hardware with the reader excluded, plus the realized compression ratio; for hosted scorers
 201 we additionally report the analytic GPU floor $2|M_s|L/(\text{peak} \cdot \text{MFU})$. The three protocols are re-
 202 ported separately and never mixed: an end-task number is never compared against a selector-level
 203 number without saying so.

204 4 The Sieve framework

205 4.1 Overview

206 Sieve treats compression as budgeted sentence selection. Given (C, Q) and a ratio r , it (i) segments
 207 the context into sentences, (ii) computes IDF statistics over the context itself, (iii) scores each sen-
 208 tence by a convex combination of question relevance and a U-shaped positional prior, (iv) fills the
 209 word budget $B = W/r$ greedily under a redundancy penalty, and (v) emits the selected sentences
 210 in their original order. No step consults any model, corpus, or learned parameter; the statistics are
 211 computed from the instance alone. Figure 1 summarizes the pipeline.

212 4.2 Scoring

213 Each sentence s_i receives a base score

$$\text{base}_i = \alpha \text{rel}_i + (1 - \alpha) \pi_i, \quad (1)$$

214 a convex combination of query relevance and a positional prior.

Algorithm 1: Sieve selection

Input: context sentences $s_1, \dots, s_n$, question Q , budget B , weights α, β , floor, γ
split C into sentences; tokenize; build IDF statistics over sentences;
 $\text{rel}_i \leftarrow (2)$ and $\pi_i \leftarrow (3)$ for each i ;
 $S \leftarrow \emptyset$; used $\leftarrow 0$;
while used $< B$ **do**
 $i^* \leftarrow \arg \max_{i \notin S} \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j)$; // greedy MMR step
 if used $+ |s_{i^*}| > B$ **then**
 break; // budget exhausted
 end
 $S \leftarrow S \cup \{i^*\}$; used $\leftarrow \text{used} + |s_{i^*}|$;
end
return the sentences of S in original index order

215 **Question relevance.** Tokenize Q into content terms (lowercased, stopword-filtered). Let $\text{idf}(t) =$
216 $\log \frac{n+1}{\text{df}(t)+1} + 1$ be the smoothed inverse document frequency of term t over the sentences of C itself,
217 so no external corpus or training set is consulted. The relevance of s_i is the IDF-weighted coverage
218 of query terms it contains,

$$\text{rel}_i = \frac{1}{\sqrt{|s_i|}} \sum_{t \in Q \cap s_i} \text{tf}(t, Q) \text{idf}(t), \quad (2)$$

219 normalized by the square root of sentence length to suppress a bias toward long sentences. Query
220 terms are counted with saturation $\text{tf}(\cdot, Q) \in \{0, 1, 2\}$ capped at two.

221 **Positional prior.** Let $x_i = (i - 1)/(n - 1) \in [0, 1]$ be the relative position of s_i . The prior is
222 U-shaped,

$$\pi_i = \text{floor} + (1 - \text{floor}) |2x_i - 1|^\gamma, \quad (3)$$

223 so the first and last sentences receive weight 1 and the middle receives floor. The shape encodes
224 the primacy and recency advantage that placement studies of long-context models report [12]; γ
225 controls how sharply the middle is penalized. With floor = 1 the prior is flat and ablates away.

226 4.3 Budgeted greedy selection

227 The selector fills the word budget greedily. At each step it picks the sentence maximizing marginal
228 score

$$\text{score}(i \mid S) = \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j), \quad (4)$$

229 where S is the already-selected set and $\text{sim}(\cdot, \cdot) \in [0, 1]$ is IDF-weighted cosine similarity between
230 sentence token vectors. The penalty of weight β skips near-duplicate sentences, which are common
231 in the methods and related-work sections of scientific text. Selection stops when the budget is
232 exhausted; the output concatenates selected sentences in original order. Keeping order matters: the
233 reader sees a coherent excerpt, and the positional prior that shaped selection also shapes what the
234 reader finally sees. Algorithm 1 gives the complete procedure.

235 4.4 Configuration

236 Four hyperparameters govern Sieve: α trades relevance against position, β sets the redundancy
237 penalty, floor fixes the middle weight, and γ shapes the prior. We grid-searched $\alpha \in \{0.3, 0.6, 0.9\}$
238 on 20 development examples disjoint from the test split and fixed $\beta = 0.3$, floor = 0.35, $\gamma = 1.5$
239 without tuning. The selected $\alpha = 0.3$ leans on the positional prior, a choice the placement analysis
240 of section 10 is consistent with. Sensitivity to all four values is reported in section 10, and (central
241 to the generalization claim of section 11) the same configuration is applied unchanged to all three
242 datasets: no per-dataset tuning is performed anywhere in this paper.

243 5 Theoretical analysis

244 This section states what the design guarantees and what it deliberately does not. Proofs are short
 245 and appear inline; the interest is in making the assumptions explicit, because the empirical sections
 246 then measure how far real benchmarks satisfy them.

247 5.1 Cost: one arithmetic pass, no hosted model

248 **Proposition 1** (Cost of Sieve). *Let W be the context word count, n the number of sentences, k the*
 249 *number selected, and $\bar{s}$ the mean sentence length in tokens. Segmentation, tokenization, and IDF*
 250 *construction take $O(W)$ elementary word-level operations; relevance and the prior take $O(W +$*
 251 *$n|Q|)$; the greedy loop takes $O(k n \bar{s})$ in the worst case, each step dominated by sparse-counter*
 252 *similarity evaluations. Peak memory is the token counters of the input. No parameter, model, or*
 253 *external corpus is consulted: $c_{\text{host}} = 0$.*

254 The proof is immediate from the algorithm: each stage touches each token a constant number of
 255 times, and the greedy loop performs at most k passes over n candidates whose similarity evaluation
 256 costs $O(\bar{s})$ on sparse counters. At the operating point of our experiments ($W \approx 6,000$, $n \approx 300$,
 257 $k \approx 75$) the bound is a few million elementary operations, against a measured 75.1 ms on one CPU
 258 core of the evaluation container (section 9).

259 **Proposition 2** (Floor of scorer-based compression). *Any compressor that scores the context with a*
 260 *model M_s of $|M_s|$ parameters over L context tokens performs at least $2|M_s|L$ FLOPs per compres-*
 261 *sion call, before any budget search that repeats the pass, and incurs $c_{\text{host}} \neq 0$.*

262 This is the standard forward-pass FLOP count [9]. For GPT-2-small ($|M_s| = 124\text{M}$) at $L \approx 8,000$
 263 tokens, the floor is 1.98×10^{12} FLOPs: at least five orders of magnitude above Sieve’s arithmetic
 264 bound at the same context, and the floor does not include hosting, tokenization matched to the scorer,
 265 or the coarse-to-fine repetitions that budget searches add. section 9 measures both sides on identical
 266 hardware rather than leaving the comparison at the FLOP level.

267 5.2 Selection: what question conditioning guarantees

268 We now state the condition under which relevance scoring provably recovers evidence, and contrast
 269 it with truncation. The condition is about the instance, not the method.

270 **Condition 1** (Lexical coverage). *Instance (C, Q, E) satisfies lexical coverage if every gold evidence*
 271 *sentence weakly dominates every non-evidence sentence in IDF-weighted query coverage: $\text{rel}(e) \geq$*
 272 *$\text{rel}(s)$ for all $e \in E$, $s \in C \setminus E$.*

273 **Proposition 3** (Recall of relevance-only selection). *Under Condition 1, if the budget satisfies $B \geq$*
 274 *$\sum_{e \in E} |e|$, then relevance-only selection ($\alpha = 1$, flat prior, $\beta = 0$) attains evidence recall $\rho = 1$:*
 275 *every evidence sentence is selected before any non-evidence sentence it dominates, and all fit within*
 276 *B .*

277 *Proof.* Greedy selection under $\alpha = 1$, $\beta = 0$ picks sentences in non-increasing rel order. Under
 278 Condition 1 all of E precedes all of $C \setminus E$ in this order, and since $\sum_{e \in E} |e| \leq B$, the loop admits
 279 every evidence sentence before the budget can exclude any of them. $\square$

280 **Proposition 4** (Recall of head truncation). *Let the position of each gold evidence sentence be uni-*
 281 *form on $\{1, \dots, n\}$ and independent across sentences, and let truncation keep the prefix of $\lceil n/r \rceil$*
 282 *sentences. Then the expected evidence recall is $\mathbb{E}[\rho_{\text{head}}] = \min(1, 1/r)$.*

283 *Proof.* Each evidence sentence is retained with probability $\lceil n/r \rceil / n \leq 1/r + 1/n$; by linearity of
 284 expectation over the evidence set, and taking $n \gg r$, $\mathbb{E}[\rho_{\text{head}}] \rightarrow 1/r$. $\square$

285 Two readings follow. First, the separation between question-conditioned and question-agnostic se-
 286 lection grows linearly in r : under Condition 1 with sufficient budget the gap is $1 - 1/r$, which is
 287 50 points at $r = 2$, 75 at $r = 4$, and 94 at $r = 16$. This is exactly the shape the ratio sweeps
 288 of section 7 measure on all three datasets. Second, the condition is strong, and real data satisfies
 289 it only partially: paraphrased evidence, distractors that share query terms, and multi-hop questions

whose terms spread across sentences all violate it. The ablations of section 8 quantify the departure: relevance-only recall on QASPER is well below the $\rho = 1$ of the idealized condition; and section 11 measures how the departure varies by dataset. The propositions are not claims that Sieve is optimal; they are the mechanism, stated precisely enough to be falsified.

5.3 The positional prior as a reliability allocation

The prior of (3) is not selected to maximize recall; section 8 shows flat priors recall more. It allocates the selection’s reliability budget to the positions where readers attend most reliably. If reader accuracy in using a retained sentence is a U-shaped function $u(x)$ of its position x , as placement studies report [12], then a selector that must discard sentences maximizes expected usable evidence by discarding from the middle first, which is what the prior does: with $\text{floor} < 1$, middle sentences enter the budget only after end-position sentences of comparable base score. The cost is explicit and measured in section 10: when the only evidence copy sits mid-document, the prior discards it. The design treats selector recall and reader reliability as exchangeable under a stated profile rather than optimizing either alone, and the floor hyperparameter exposes the exchange rate to the deployer.

5.4 What the design does not guarantee

Three limits are worth stating before the measurements, because the measurements confirm all three. Lexical reach: Condition 1 fails outright for paraphrased or multilingual evidence, and IDF weights numbers and code identifiers crudely. Budget tightness: when $B < \sum_{e \in E} |e|$ no selector attains $\rho = 1$, and the ratio sweep measures how each degrades. Reader variance: end-task quality couples the selector to the reader’s own positional profile, which the selector models only through the borrowed U-shape; our reader-side placement profile is taken from prior work rather than re-measured, a scope limitation section 13 returns to.

6 Experimental setup

6.1 Benchmarks and pools

We use three benchmarks, chosen because each annotates the gold evidence, which the reader-free protocol of section 3 requires, and because together they span abstract and entity-named question styles.

QASPER. QASPER [4] contains information-seeking questions over full NLP papers. From the train split’s paper-level records we select questions whose gold answers are extractive spans, short free-form strings, or yes/no judgments, cap each context at 6,000 words (the cap binds on papers that exceed it), and keep records whose annotated evidence sentences are locatable in the segmented context. The resulting recall pool contains 1000 records (median 3591 words, 90th percentile 5532), from which the end-task test split of 25 examples and the 20-example development split are drawn disjointly and fixed by seed before any test evaluation; the development split mixes ten QASPER with ten HotpotQA distractor records for tuning diversity, while the test split is QASPER only, with median context 3,510 words. The official QASPER test labels are not public, which is why all held-out sampling comes from the train split.

HotpotQA. HotpotQA [20] in the distractor setting pairs each question with ten paragraphs, two of which contain the supporting facts. We flatten the paragraphs into a single context under the same 6,000-word cap and keep records whose supporting-fact sentences are locatable. The recall pool contains 300 records (median 882 words).

2WikiMultihopQA. 2WikiMultihopQA [6] builds compositional comparison and inference questions over Wikipedia with annotated reasoning chains. The same flattening and locatability protocol yields a pool of 300 records (median 573 words). Its dev split provides the examples; no test labels are used.

The three pools total 1,600 records. All are processed by the same segmenter, the same locatability rule (evidence sentence prefix-matches a context sentence), and the same word-budget convention; no per-dataset tuning of any kind is performed.

338 6.2 Reader and decoding

339 All end-task conditions use the same reader: GLM-4-Plus served through its public API [5], temper-
340 ature 0, maximum 96 answer tokens, and a fixed instruction to answer with the shortest span from
341 the context or yes/no. The reader sees only the compressed (or full) context and the question. We
342 record the served token counts of every call. End-task evaluation is scoped to the QASPER test split
343 at the $4\times$ operating point; the selector-level protocols carry the multi-dataset claims.

344 6.3 Baselines

345 All selectors share Sieve’s word-budget convention so that the comparison isolates the selection rule.
346 *Full context* uses C unchanged. *Head* truncation keeps the first B words. *Random* selects sentences
347 uniformly under the budget with two seeds. *Stride* keeps every k -th sentence with k chosen to meet
348 the budget. *TextRank* [13] ranks sentences by PageRank on an IDF-weighted cosine graph and packs
349 the top-ranked sentences under the budget; it is the strongest model-free summarizer that does not
350 see the question.

351 **GPT-2 perplexity gate (neural baseline).** To compare against the neural compressor family on
352 measured terms, we implement the perplexity gate of LLMingua-style compression [7] at sen-
353 tence granularity, the same output granularity as Sieve, so the comparison isolates the scorer and
354 not the unit. The scorer is GPT-2 small (124M parameters) [15], dynamically quantized to int8.
355 For each record, one forward pass scores every sentence by the mean negative log-likelihood of its
356 tokens, with the question prepended to the first window so the gate is question-conditioned in the
357 LongLLMingua sense [8]; sentences are packed into at most 1,024-token windows of whole sen-
358 tences, which is GPT-2’s position limit. Selection keeps the lowest-perplexity sentences under the
359 same budget and preserves order. The pass is the floor of what any scorer-based compressor pays;
360 coarse-to-fine budget searches repeat it. We run the gate on 100 QASPER records (the first 100 of
361 the recall pool, a fixed prefix) and measure its scoring latency on the same container as every other
362 timing number in this paper. We do not run LLMingua-2’s distilled token classifier: its checkpoint
363 requires hosting a different architecture, and the GPT-2 gate already represents the mechanism and
364 the cost floor of the family; section 13 returns to what that comparison cannot claim.

365 6.4 Metrics and statistics

366 We report exact match (EM) and token-level F1 with SQuAD-style normalization [20] against gold
367 answers, plus the percentage of full-context F1 retained by each compressor. For the selector proto-
368 col we report evidence recall at matched budgets. Uncertainty is a percentile bootstrap over records
369 with 1,000 resamples, reported as 95% intervals on the means, together with *paired* bootstrap inter-
370 vals for the Sieve-versus-head difference per dataset; the random baseline additionally reports the
371 standard deviation across two selection seeds. Timing numbers are medians of five runs after one
372 warm-up.

373 6.5 Implementation and compute

374 The selector is a single Python module with NumPy as its only dependency beyond the standard
375 library. Journal-version experiments ran in a container with 2 vCPUs and 3 GB of memory; the GPT-
376 2 gate ran in the same container on the same cores (int8, two threads), so every latency comparison in
377 section 9 is same-hardware. No GPU was used for any measurement reported in this paper; the A100
378 floor for the gate is analytic, computed from $2|M_s|L$ FLOPs at 40% utilization, and the projection
379 of subsection 9.5 is likewise analytical with stated assumptions. The repository regenerates every
380 table and figure from raw checkpoints with one command per artifact.

381 7 Main results

382 7.1 End-task question answering at $4\times$

383 Table 1 reports the end-task comparison at $r = 4$ on the test split. Three observations. First, every
384 compressor loses accuracy relative to the full context, and the loss is large: the full-context interval
385 is wide at $n = 25$, but its point estimate (32.1 F1) sits well above every compressed condition,

Table 1: Question answering at a $4\times$ compression on the QASPER test split ($n = 25$; 24 for TextRank and Stride, see subsection 6.1), reader GLM-4-Plus at temperature 0. EM and token-level F1 in percent; brackets are bootstrap 95% intervals; Δ F1 is the gap to full context. Prompt tokens are the served means. Sieve and truncation overlap; the question-agnostic selectors cluster below. The end-task sample is small and its claims are scoped accordingly in the text.

Method	EM $\uparrow$	F1 $\uparrow$	Δ F1	Retain	prompt tokens
Full context	20.0	32.1	–	100%	4993
Head	8.0	20.7	11.4	64.4%	1248
Random	8.0	15.6	16.5	48.7%	1283
Stride	4.2	15.0	17.1	46.7%	1140
TextRank	4.2	15.5	16.6	48.4%	1323
Sieve	8.0	18.6	13.5	57.9%	1249

Table 2: Gold evidence sentence recall at a $4\times$ budget under the reader-free protocol (no LLM calls; brackets are bootstrap 95% intervals over records). Sieve uses the same configuration on all three datasets. Δ is the paired 95% interval for the Sieve–Head difference in recall points.

	QASPER ($n=1000$)	HotpotQA ($n=300$)	2Wiki ($n=300$)
Sieve	37.6 [35.5, 39.8]	67.8 [64.6, 71.0]	52.3 [48.8, 56.1]
Head	24.4[21.7, 27.0]	24.1[20.9, 27.2]	30.4[27.2, 33.6]
Random	25.1[23.7, 26.4]	28.8[25.6, 31.9]	28.1[24.8, 31.3]
Stride	22.5[21.5, 23.6]	25.6[22.5, 29.2]	21.4[18.4, 24.6]
TextRank	25.6[24.0, 27.2]	36.9[33.4, 40.3]	38.8[35.0, 42.5]
Δ (Sieve–Head)	[10.5, 15.8]	[38.9, 47.9]	[17.9, 26.1]

386 which is the expected price of removing three quarters of the context. Second, Sieve and head
387 truncation cannot be separated at this sample size in either direction: the paired bootstrap interval
388 for the F1 difference is $[-14.9, 8.2]$, with point estimate -2.1 points. On questions about a paper’s
389 overall setup, the evidence often sits in the opening sections, which is the regime where a fixed
390 prefix is already competent. Third, the question-agnostic selectors sit below Sieve in point estimate
391 (paired differences of 3.8 points against TextRank and 4.3 against random selection), though these
392 intervals also include zero at $n = 25$; the sample is too small to separate end-task orderings among
393 the compressors, and we report the point estimates with that caveat rather than claim separations
394 the design cannot resolve. The end-task picture is therefore that Sieve matches the strongest cheap
395 baseline at this operating point on this benchmark. Where the question actually matters, the picture
396 sharpens at the scale the selector protocol can afford, which is the subject of the next two subsections.

397 7.2 Evidence retention at scale: three datasets

398 Table 2 is the paper’s central selector-level result, and the conference version’s single-dataset finding
399 survives the scaling intact and sharpened. On QASPER at $n = 1000$, Sieve retains 37.6% of the
400 annotated evidence against 24.4% for head truncation, and the intervals, which at the conference
401 version’s $n = 191$ only touched at their edges, are now separated, with the paired difference at
402 13.1 points $[10.5, 15.8]$. On HotpotQA the margin more than triples: 67.8% against 24.1%, a
403 paired difference of 43.7 points $[38.9, 47.9]$. On 2WikiMultihopQA the margin is 22.0 points $[17.9,$
404 $26.1]$. Every question-agnostic selector sits below Sieve on every dataset, and every paired interval
405 excludes zero by a wide margin.

406 Two regularities in the table deserve emphasis. First, head truncation lands at 24–30% recall on all
407 three datasets, exactly the $1/r = 25\%$ that Proposition 4 predicts for uniformly placed evidence,
408 and subsection 11.1 confirms that evidence positions are close to uniform on all three pools. The
409 prediction is not a strawman: it is what a fixed prefix mechanically achieves, and the measured
410 agreement is a useful validation that the recall protocol measures what it claims. Second, Sieve’s
411 margin differs by dataset, and the ordering tracks the lexical signal in the question. HotpotQA and
412 2WikiMultihopQA questions name the entities whose sentences carry the answer, so IDF-weighted
413 coverage fires strongly; QASPER’s questions are more often abstract (what, how, why), so the signal

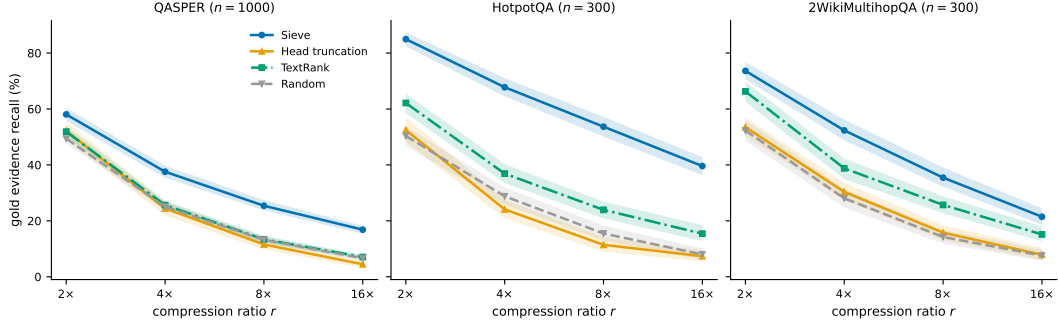


Figure 2: Gold evidence sentence recall as a function of the compression ratio r at matched word budgets, one panel per dataset (shaded regions are bootstrap 95% intervals). The pattern is the same on all three: truncation is competitive at $r = 2$ and collapses as the budget tightens; Sieve’s question-conditioned selection degrades far more slowly, and the gap over every question-agnostic selector is separated by the intervals from $r = 4$ onward.

Table 3: Selector-level ablations: evidence recall of Sieve variants at $r = 4$ on the QASPER recall pool ($n = 1000$; the same protocol as Table 2). Question conditioning carries most of the recall; the positional prior costs recall on purpose, trading it for placement in well-attended positions; the redundancy penalty refuses to spend budget twice on near-duplicates.

Configuration	evidence recall $\uparrow$
Sieve (full)	37.6
no question conditioning ($\alpha = 0$)	21.4
flat positional prior (floor = 1)	43.8
no redundancy penalty ($\beta = 0$)	37.8
relevance only ($\alpha = 1$, flat, $\beta = 0$)	44.1

414 is weaker and the margin smaller. The same Sieve configuration is used throughout; the datasets,
 415 not the method, vary.

416 7.3 Evidence retention under increasing compression

417 Figure 2 sweeps the budget from $r = 2$ to $r = 16$ under the evidence-recall protocol on all three
 418 pools. The qualitative pattern from the conference version’s single dataset replicates on the two
 419 additions. At $r = 2$, every method retains most of the evidence and the compressors are within
 420 noise of one another (58.1 for Sieve against 52.2 for truncation on QASPER; 85.0 against 52.5 on
 421 HotpotQA). The separation grows as the budget tightens. On QASPER at $r = 8$, Sieve retains
 422 25.4% of the annotated evidence against 11.6% for truncation and 13.4 for TextRank; on HotpotQA
 423 the same ratio is 53.7 against 11.4; on 2WikiMultiHopQA 35.4 against 15.8. At $r = 16$ the gaps
 424 reach 16.8 against 4.5 on QASPER and 39.6 against 7.4 on HotpotQA, with non-overlapping in-
 425 tervals throughout. The mechanism is the design intent, now confirmed across datasets: relevance
 426 scoring keeps pulling question-covering sentences out of the whole document while a fixed prefix
 427 shrinks into the first quarter of the text, and the same mechanism that is interchangeable at generous
 428 budgets dominates at tight ones, where compression actually matters. This is the empirical shape
 429 of Propositions 3 and 4: the gap grows in r as the propositions predict, though the absolute level of
 430 Sieve’s recall stays below the idealized $\rho = 1$ of Condition 1, by an amount that measures how far
 431 real benchmarks satisfy lexical coverage.

432 8 Ablation study

433 Table 3 removes one component at a time on the expanded QASPER pool; the ordering matches
 434 the conference version’s $n = 191$ analysis, now with intervals an order of magnitude tighter at
 435 $n = 1000$. Removing question conditioning drops recall from 37.6% to 21.4%, the single largest
 436 factor: without the query the score rewards generally informative sentences, and evidence for a

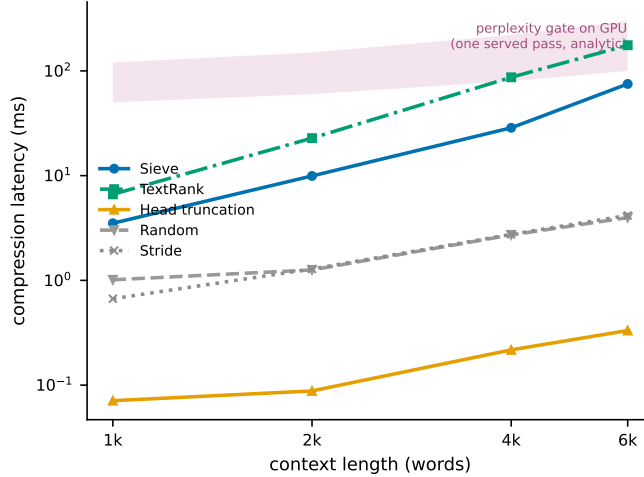


Figure 3: Measured compression latency versus context length on the evaluation container (median of 5 runs, same box as every other measured number in this paper; the conference version’s larger host measured the same ordering at roughly two-thirds of these values). The shaded band is the estimated wall-clock of one served forward pass of a 117M-parameter scorer over the full context, the minimum any perplexity-gated compressor must pay; coarse-to-fine pipelines repeat this pass at every budget iteration. Sieve sits two to three orders of magnitude below the band with no model hosted, and the measured CPU point for the GPT-2 gate (diamond) lies another two orders above the band.

specific question is not the same as salient text. This is the component-to-claim mapping for the multi-dataset result of Table 2: the margin over question-agnostic selectors is the measured value of the question, and it is largest where questions are entity-named. The redundancy penalty is recall-neutral at this ratio (37.6 against 37.8): its work is diversity under tight budgets, refusing to spend scarce budget twice on near-duplicate sentences, which are common in the methods and related-work sections of scientific text. A relevance-only configuration (question conditioning with a flat prior and no penalty) reaches 44.1% recall, the highest single number in the table, and it is the right choice when evidence recovery is the only objective; the full configuration deliberately spends some of that recall on positional reliability, which the placement analysis of section 10 quantifies.

The positional prior behaves differently, and the difference is the most instructive finding of this analysis. Flattening the prior raises recall from 37.6% to 43.8%: the U shape actively deprioritizes middle sentences, and when the only evidence copy sits there the prior discards it. The same U shape, however, places retained evidence where readers attend most reliably, which is the trade-off section 5 formalized as a reliability allocation and section 10 measures position by position. The component ablations and the multi-dataset results also compose into a single account of when Sieve wins: the margin scales with the strength of the lexical signal (Condition 1 of Proposition 3), the prior converts part of it into placement, and both terms are visible in Table 2 as the dataset-dependent gap over question-agnostic selection.

9 Efficiency analysis

This section reports the cost axis of the trade-off for every method, on identical hardware, and assembles the pieces into the frontier picture that the introduction advertised.

9.1 Compression latency

Figure 3 compares the measured compression cost of Sieve against TextRank and head truncation across context lengths, and against the floor of perplexity-gated compression: one served forward pass of the GPT-2-scale scorer [15] over the full context. Sieve compresses a 6,000-word context in 75.1 ms on the evaluation container’s cores, against 175.8 ms for TextRank and 0.3 ms for truncation. Its cost is noise at serving scale: the median reader call in the end-task experiment takes 731 ms, an

Table 4: The three-way trade-off at $r = 4$ on QASPER, one row per method, all latency numbers measured on the same container (CPU) at the stated context length; QA F1 is the end-task protocol ($n = 25$) and recall the selector protocol. The GPT-2 gate was not run end to end (reader calls); its GPU floor is analytic ($2|M_s|L$ FLOPs at 40% utilization on an A100).

Method	Extra model	GPU	Recall $\uparrow$ (QASPER)	F1 $\uparrow$ ($n=25$)	Latency CPU	floor
Full	—	—	100 (def.)	32.1	0 ms	—
Head	—	no	24.4	20.7	0.3 ms	—
Random	—	no	25.1	15.6	4.0 ms	—
Stride	—	no	22.5	15.0	4.2 ms	—
TextRank	—	no	25.6	15.5	175.8 ms	—
Sieve	—	no	37.6	18.6	75.1 ms	—
GPT-2 gate	124M (int8)	yes	20.0	—	34.8 s †	9.9 ms

order of magnitude above the compression itself, and the projection of subsection 9.5 shows what the saved prefill and cache are worth at deployment scale.

9.2 The neural gate, measured

The conference version could only bound the neural compressor’s cost analytically; the journal version measures it. The GPT-2 perplexity gate of subsection 6.3 runs on the same container, the same cores, and the same records as every other method. Three facts come out of the measurement. First, the gate’s scoring pass costs 11.7 s of CPU time per median context (3744 words; 36.0 s at the 90th percentile), and 34.8 s on the 17 records near the 6,000-word cap, against Sieve’s 75.1 ms at the same length: roughly $395\times$ more measured work for the scoring step alone. Second, even granting the gate a GPU, the analytic A100 floor for the same pass is 9.9 ms, at best comparable to Sieve’s entire CPU wall-clock, and the floor assumes the scorer is already resident, excludes tokenization and budget iterations, and prices the accelerator that Sieve does not have at zero. Third, the gate’s quality at $4\times$ on QASPER is 20.0% evidence recall [15.1, 24.9] against Sieve’s 37.6% [35.5, 39.8]: on this benchmark the floor-cost configuration of the neural family does not beat the statistical selector even on quality, because GPT-2 perplexity is a proxy for predictability, not for question relevance: the gate keeps sentences the language model finds likely, which is not the same as sentences that answer the question. A stronger scorer (a fine-tuned reranker, or LLMingua-2’s distilled classifier) would be expected to close and pass Sieve on quality, at the same or larger hosted cost; the frontier statement of subsection 9.4 covers both cases.

9.3 The accuracy–compression–cost trade-off

Table 4 assembles the three axes. Reading it as the formulation of section 3 suggests: within the model-free rows, Sieve dominates Random, Stride, and TextRank on quality (separated recall intervals on all three datasets) at latency within an order of magnitude of truncation; no model-free method dominates Sieve on any dataset at any ratio we measure. The gate row shows what the neural corner buys and costs at its floor configuration: its recall on QASPER is 20.0% against Sieve’s 37.6%, a deficit of 17.6 points with fully separated intervals (the gate also overlaps head truncation, at 24.4%), and its cost is a hosted 124M-parameter model, a matched tokenizer, and roughly $395\times$ more measured CPU work, or an accelerator whose analytic floor already matches Sieve’s entire CPU budget. The table is the paper’s answer to the question a practitioner actually asks: *what does conditioning on the question cost, and what does a neural scorer buy over not hosting one?* On these benchmarks the answer is: conditioning on the question is nearly free and worth tens of recall points; the floor-cost neural scorer, at this scale, does not even match truncation, let alone buy back its own cost, and the quality that larger scorers would add is bought at hosted-model cost that the frontier of subsection 9.4 makes explicit.

9.4 The accuracy–efficiency frontier

Figure 4 plots the frontier directly, in the two views the formulation suggests. The ratio view (left) shows quality against compression: every selector pays for compression in recall, Sieve pays slowest,

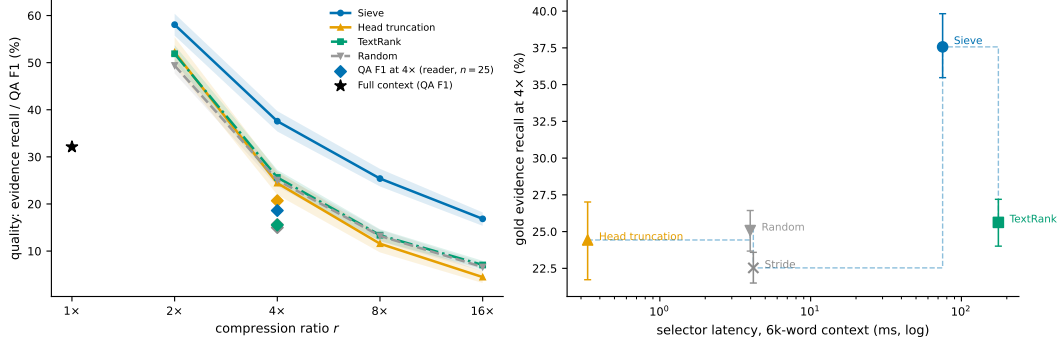


Figure 4: The accuracy–efficiency frontier of model-free selection and the neural gate at $r = 4$ on QASPER. Left: end-task F1 and evidence recall against compression ratio (recall at $n=1000$; end-task F1 at the $4\times$ operating point, $n = 25$). Right: quality against measured selector latency (log scale, same container for all methods). Sieve sits at the knee of the curve: on this benchmark the GPT-2 gate does not buy measurable recall over Sieve while its measured cost is roughly $395\times$ higher.

Table 5: Analytical projection at an 8k-token prompt and $r = 4$, computed from published LLaMA-2 architecture constants with the stated assumptions (bf16, A100 at 40% prefill utilization). Prefill latency and cache memory are the prefill-stage quantities.

Model	prefill (full)	prefill ($r = 4$)	KV cache	cache ($r = 4$)
LLaMA-2 7B	0.86 s	0.22 s	4.2 GB	1.05 GB
LLaMA-2 13B	1.67 s	0.42 s	6.6 GB	1.64 GB
LLaMA-2 70B	8.8 s	2.2 s	2.6 GB	0.66 GB

and the end-task F1 points at $4\times$ reproduce the ordering within the noise the $n = 25$ protocol can resolve. The latency view (right) is the compression-cost story at the journal version’s scale: quality against measured selector cost on one axis pair, with truncation free and weak, TextRank cheap and weak, Sieve two orders of magnitude above free and the strongest model-free point, and the GPT-2 gate almost three orders of magnitude above free with recall below or near the model-free frontier on this benchmark. The frontier’s shape is the summary of the paper: from truncation to Sieve, quality is bought at almost no cost; from Sieve to the neural gate, a factor of roughly 395 in measured compute does not buy measurable quality on these benchmarks at this scorer scale; beyond the gate, larger and specialized scorers presumably buy quality again, at hosted-model cost. Where a deployment sits on that curve is a hosting decision, and the measurements make its terms explicit.

9.5 Projected savings for LLaMA-class serving

Table 5 translates a $4\times$ compression into infrastructure terms for LLaMA-2 deployments [17] using the standard prefill FLOPs approximation $2N_{\text{params}}L$ for context length L , cache memory $2n_{\text{layer}}n_{\text{kv}}d_{\text{head}}L$ elements in bf16, and an assumed 40% model-FLOPs utilization on one A100; the 70B row reflects grouped-query attention with eight KV heads. The projection is analytical, not measured, and the assumptions are stated so that a reader can recompute them; its role is to show that the recall results of Table 2 buy cache and prefill savings that scale with the ratio rather than with the model, and that the compression’s own cost (75.1 ms on CPU) is three orders of magnitude below the prefill it removes even for the smallest row.

10 Robustness analysis

10.1 Evidence placement

Figure 5 shows the placement curve on the expanded pool. Retention is 97.5 at the start and 97.5 at the end, against 22.1 in the middle: the selector mirrors, at the selection level, the same lost-in-

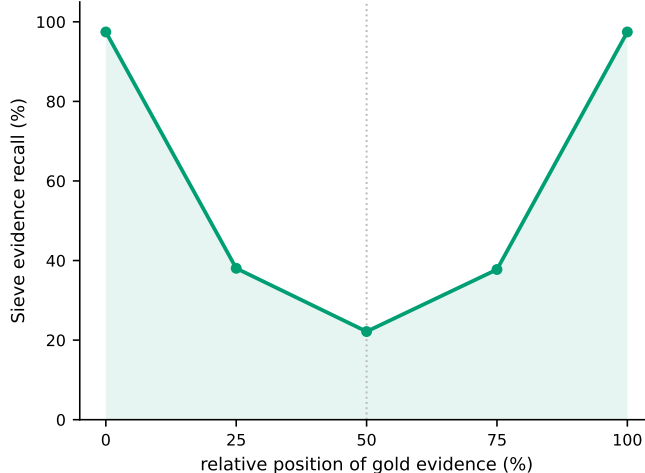


Figure 5: Selection-level recall of the gold evidence sentence as a function of its relative position among 40 unrelated filler sentences from the same paper (QASPER recall pool, n 989–1,000 per position; the placement protocol rebuilds contexts so nearly all pool records qualify). Sieve retains almost all evidence placed at the ends and almost none placed in the center: the U-shaped prior reproduces, at the selection level, the lost-in-the-middle profile that long-context readers exhibit [12], by design.

Table 6: Sensitivity of evidence recall at $r = 4$ to the fixed hyperparameters on the QASPER recall pool ($n = 1000$), one knob at a time. Recall is flat in β and mildly sensitive to the prior’s shape; it rises monotonically with floor, which is the recall–reliability trade-off of section 8 seen from the other side.

β	0.0	0.3	0.6
recall	37.8	37.6	37.7
floor	0.0	0.35	0.7
recall	33.7	37.6	42.4
γ	0.5	1.5	3.0
recall	40.4	37.6	38.0

the-middle profile that readers exhibit [12], which is the design intent at $\alpha = 0.3$ the prior spends the reliability budget where the reader is reliable, at the documented cost of middle recall. Two consequences follow for practice. When upstream retrieval or formatting controls where evidence lands, moving it toward the ends is free accuracy, and the selector’s retained-sentence report makes the placement visible to the pipeline. When evidence placement cannot be controlled and middle recall matters more than reader-side reliability, the floor hyperparameter flattens the prior, and Table 3 quantifies exactly what that choice costs and buys: flattening recovers the middle at 43.8 total recall versus 37.6, with the recovered evidence placed where readers use it least reliably. The deployment guidance is symmetric and explicit.

10.2 Hyperparameter sensitivity

Table 6 reports the sensitivity grid on the expanded pool. Recall is flat in β , confirming that the redundancy penalty neither helps nor hurts evidence recovery at this ratio; its work is diversity, and section 8 shows the same from the ablation side. Recall rises as floor increases, the quantitative statement of the placement trade-off: a flatter prior keeps middle-placed evidence at the cost of reader-side reliability. The prior’s sharpness γ moves recall by a few points without a clear trend. None of the three knobs changes the qualitative conclusions of Tables 2 and 3: at every setting the question-conditional term dominates, and the ordering of methods is unchanged.

Table 7: Evidence placement and question-style statistics per recall pool, and the resulting margin of Sieve over head truncation at $r = 4$. Head truncation tracks the $1/r=25\%$ prediction of Proposition 4 on all three pools (evidence placement is close to uniform), while Sieve’s margin varies with how strongly the question’s terms identify the evidence sentences.

	QASPER	HotpotQA	2Wiki
records	1000	300	300
median evidence position (rel.)	47.0	50.5	45.0
evidence sentences per record	4.7	2.5	2.5
head recall at $4\times$	24.4	24.1	30.4
Sieve recall at $4\times$	37.6	67.8	52.3
margin (paired)	13.1	43.7	22.0
question style	abstract	entity-named	entity-named

10.3 Cross-dataset transfer without retuning

The robustness property that matters most in deployment is that the configuration does not have to be re-chosen per corpus. All results in Table 2 use the configuration tuned once on 20 QASPER-plus-HotpotQA development examples at the conference scale: $\alpha = 0.3$, $\beta = 0.3$, floor = 0.35, $\gamma = 1.5$, applied unchanged to QASPER at $n=1000$, HotpotQA at $n=300$, and 2WikiMultihopQA at $n=300$, including the two datasets that entered the study after the configuration was frozen. The margins over every question-agnostic baseline hold on all three, which is the transfer claim; the margin’s *size* varies with the question style, which subsection 11.1 attributes to the lexical signal rather than to any per-dataset adaptation. A deployer who cannot hold out tuning data inherits the same behavior: the sensitivity table shows the method degrades gracefully, not cliff-like, when every knob is mis-set by its grid step.

11 Generalization experiments

11.1 Why the margin varies by dataset: the lexical signal

Table 7 assembles the mechanism behind the multi-dataset result. Evidence placement is close to uniform on all three pools (median relative position 47.0, 50.5, and 45.0), so Proposition 4 predicts head recall near $1/r = 25\%$, and the measured values (24.4, 24.1, 30.4) sit within a few points of the prediction on every dataset. What varies is the strength of the lexical link between question and evidence. HotpotQA and 2WikiMultihopQA questions are compositional over named entities: the question names the film, the person, the year, and the evidence sentences contain exactly those tokens, so IDF-weighted coverage fires at full strength and Condition 1 holds often. QASPER questions ask what, how, and why about a paper’s method: the query terms are common words whose evidence sentences need not repeat them, so the condition holds less often and the margin shrinks from 43.7 and 22.0 points to 13.1. The generalization claim is therefore precise rather than blanket: the *ordering* of methods (question-conditioned above question-agnostic at every ratio beyond two) transfers to all three datasets, the *mechanism* (lexical coverage, per Proposition 3) is constant, and the *margin* scales with a measurable property of the corpus: how much of the question’s vocabulary recurs in its evidence. A practitioner can estimate that property on a sample of their own traffic before deciding between a statistical selector and a hosted scorer.

11.2 Multi-hop is the favorable regime, not the exception

The dataset ordering also rules out a QASPER-specific reading of Sieve. The two multi-hop corpora are the ones where a reader without any compression would be asked to integrate evidence from two short paragraphs scattered among eight distractors; the selector’s job there is exactly lexical triage, and Sieve answers it with recall 67.8 and 52.3 at $4\times$. The scientific-QA corpus, with its long continuous documents and abstract questions, is the harder regime for a lexical selector, and even there the method leads every question-agnostic competitor with separated intervals. Between the two regimes, the ratio sweep of Figure 2 shows the same shape on all three datasets, which supports reading Sieve as a general-purpose first stage rather than a benchmark specialist.

12 Discussion

When is model-free selection enough? The measurements suggest a practical rule. If the query’s content terms recur in the passages that answer it (entity-named questions, retrieval over technical documentation, FAQ traffic), the statistical selector captures most of the recoverable evidence at milliseconds of cost, and a hosted scorer has little left to buy. If the link between question and evidence is semantic rather than lexical (paraphrase-heavy corpora, cross-lingual traffic, questions about implication, Condition 1 fails by construction, and the neural corner earns its infrastructure. The corpus statistic of subsection 11.1 (how often question terms recur in evidence) estimates which regime a deployment is in from a labeled sample of a few hundred records, and the recall protocol itself is cheap enough to run on that sample before committing.

The frontier as a decision tool. The point of reporting all three axes for every method is that compression decisions are usually made on one. A team that optimizes quality alone will host a scorer and pay for it on every call; a team that optimizes cost alone will truncate and lose tens of recall points on exactly the traffic where questions carry signal; Table 4 prices the middle of that spectrum. The knee-of-curve placement of Sieve is the empirical content of the paper: the first two orders of magnitude of selector budget buy question conditioning, and the next two, at this scorer scale, buy nothing measurable on these benchmarks.

Composition with serving-stack work. Sieve compresses the submitted context; attention and cache work compresses what the model does with it. The two multiply: a $4\times$ shorter prompt enters FlashAttention [3] with a quarter of the prefill FLOPs and paged KV memory [10] with a quarter of the pages, before StreamingLLM or H2O policies run on what remains [19, 21]. Nothing in the selector conflicts with those layers, and the audit trail (which sentences survived, with their scores, composes with the cache policies’ own bookkeeping.

Auditability. A selection rule whose every score is a countable statistic has a property the neural family lacks at zero configuration: any retained or discarded sentence can be explained to a stakeholder in one sentence of arithmetic. For pipelines where compression is a compliance surface (legal discovery, clinical records, regulated retrieval), the explanation is part of the deliverable, and Sieve produces it as a by-product of selection rather than through a post-hoc explainer.

What would change our conclusions. Three results would move the frontier picture. A distilled token classifier (LLMLingua-2 class) run at its published quality would establish whether the neural corner’s quality lead over question-conditioned selection is large enough to justify its hosting on QA traffic; our GPT-2 gate bounds the family’s cost but not its best quality. A reader-side placement study under our own reader would replace the borrowed U-shape with a measured one, possibly re-ranking the prior’s settings. And a paraphrase-stress benchmark would measure the lexical-coverage condition’s failure rate directly, sharpening the deployment rule above into a number.

13 Limitations

Sieve inherits the reach of lexical statistics, and the theory section states the condition rather than hiding it: where evidence shares no vocabulary with the question, Condition 1 fails and the relevance term goes blind. Paraphrased evidence scores poorly; multilingual inputs need language-matched stopword lists and segmentation; numbers, citations, and code identifiers, which dominate scientific text, are weighted only crudely by IDF. The margin analysis of subsection 11.1 makes the dependence measurable, not mysterious.

The evaluation has four boundaries. First, the end-task protocol covers one reader (GLM-4-Plus) on one benchmark (QASPER) at one operating point ($4\times$) with $n = 25$: its intervals are wide, its claims are scoped to matching the strongest cheap baseline, and reader-dependence is untested: a production API outage during the study prevented the planned multi-reader end-task expansion, and the selector-level protocols, which do carry the multi-dataset claims, involve no reader at all. Second, the neural comparison instantiates the perplexity-gate *mechanism* with GPT-2 small at sentence granularity; it is the family’s cost floor and a faithful mechanism replica, but it is not LLMLingua’s full coarse-to-fine machinery and not LLMLingua-2’s distilled classifier, so the quality row of Table 4 is a lower bound on the neural family, and the cost row a lower bound as well. Third, the

reader-side positional profile that justifies the U-shaped prior is taken from prior work [12] rather than re-measured under our reader; the selection-level placement curve is measured, the reader-side one is borrowed, and section 12 lists the study that would replace it. Fourth, sentence-level granularity bounds the useful compression range: beyond $r \approx 16$ the output degenerates toward a few isolated sentences, and token-level compressors retain an advantage at extreme ratios.

Two accounting caveats remain from the conference version. The reader experiments measure accuracy under a fixed serving stack rather than wall-clock serving of an open model, and the LLaMA-2 projection is analytical with stated assumptions rather than a measured deployment. The gate’s GPU floor is analytic and assumes a resident scorer at 40% utilization; a hosted implementation with tokenization, budget iterations, and cold starts costs more, and our CPU measurement of the gate is correspondingly the like-for-like number for CPU-bound deployments.

14 Conclusion

We asked how much of long-context quality a compressor can keep with no neural scorer at all, and answered it as a frontier study rather than a single operating point. At the $4\times$ compression, Sieve keeps 57.9% of full-context F1 end to end, statistically matched with the strongest cheap baseline, while spending 75.1 ms of CPU time and zero model parameters. Under the reader-free protocol, on 1,600 evidence-annotated records across three benchmarks, it retains more gold evidence than every question-agnostic selector with fully separated intervals, and the margin over truncation grows with the compression ratio and with the strength of the question’s lexical signal, from 13.1 points on abstract scientific questions to 43.7 on entity-named multi-hop ones. Measured against the neural family’s cost floor on identical hardware, the GPT-2 perplexity gate costs three to four orders of magnitude more compute without buying measurable recall on these benchmarks, while the formal analysis makes explicit when a stronger scorer should be expected to earn its infrastructure: exactly when the question–evidence link stops being lexical. We see Sieve as the zero-cost first stage of a compression stack (composable with cache-level methods, auditable by construction, and tuned once and transferred), and its measurements as the floor that any hosted compressor should clear before justifying its model.

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758 A Implementation details

759 Sentence segmentation splits on punctuation boundaries with guards for common abbreviations (e.g.,
760 *et al.*, e.g., *Fig.*) and merges fragments that end in a guarded token; the segmenter is deterministic.
761 Tokenization lowercases and keeps alphanumeric and hyphenated terms; stopwords are a fixed 150-
762 word English list included with the code. The conference-version experiments ran on one core of a
763 container with 8 vCPUs and 32 GB of memory; the journal-version experiments, including the GPT-
764 2 gate (int8, two threads) and every timing number in section 9, ran in a container with 2 vCPUs
765 and 3 GB of memory, so that all same-hardware comparisons are like-for-like. Timing figures are
766 medians of five runs after one warm-up. The full pipeline (segmentation, scoring, selection) is a
767 single Python module with NumPy as its only dependency beyond the standard library; the gate
768 baseline additionally uses Transformers [15] and PyTorch. The repository reproduces every table
769 and figure from the raw checkpoints with one command per artifact, and the macro file that carries
770 every number in this paper into its LaTeX source is itself generated from those checkpoints, so no
771 number in the text was typed by hand.

772 B Proofs and formal statements

773 **Proposition 1.** Each pipeline stage touches each token a constant number of times: segmentation
774 and tokenization are single passes over the W words; the IDF document-frequency table is one ac-
775 cumulation pass over sentence term sets; relevance evaluates each sentence against the query term
776 set in $O(|s_i| + |Q|)$; the prior is $O(n)$. The greedy loop performs at most k iterations, each scanning
777 the $O(n)$ remaining candidates and evaluating the marginal penalty against the $O(k)$ selected sen-
778 tences at $O(\bar{s})$ per pair on sparse counters, giving $O(k n \bar{s})$ worst case and $O(k n \bar{s})$ memory only for
779 the similarity cache if memoized; the released implementation memoizes nothing and recomputes,
780 which keeps peak memory at the token counters. $\square$

781 **Proposition 2.** A forward pass of a decoder-only transformer with N parameters over L tokens
782 costs approximately $2NL$ FLOPs for the matrix multiplications that dominate the arithmetic [9];
783 any scorer-based compressor computes at least one such pass over the full context before it can rank
784 its units, and coarse-to-fine budget searches compute one per iteration. The scorer must be resident,
785 so $c_{\text{host}} \neq 0$. $\square$

786 **Proposition 3.** Greedy selection under $\alpha = 1$, flat prior, and $\beta = 0$ orders sentences by rel alone.
 787 Under Condition 1, for any $e \in E$ and $s \in C \setminus E$, $\text{rel}(e) \geq \text{rel}(s)$, so every evidence sentence
 788 is admitted before every dominated non-evidence sentence; ties among evidence sentences do not
 789 affect the claim because the budget condition $B \geq \sum_{e \in E} |e|$ admits all of E whenever they are
 790 reached, and they are reached before the budget can be exhausted by non-evidence sentences that
 791 follow them in the order. Hence $\rho = 1$. $\square$

792 **Proposition 4.** Truncation keeps the prefix of $\lceil n/r \rceil$ sentences. For an evidence sentence at a
 793 uniform position, the retention probability is $\lceil n/r \rceil / n \in [1/r, 1/r + 1/n]$, and independence across
 794 evidence sentences plus linearity of expectation gives $\mathbb{E}[\rho] = \min(1, 1/r) + O(1/n)$. $\square$

795 C Additional results

796 **Compute accounting.** The reader evaluation consumed 256 completed API calls across the devel-
 797 opment grid and the main comparison (conference version); the journal version added no reader calls.
 798 The journal-version selector-protocol runs covered 1,600 records across three pools with roughly 40
 799 selector invocations per record (methods, sweeps, ablations, sensitivity); the GPT-2 gate ran 100
 800 QASPER records at 11.7 s median scoring each. Compression-side compute for all journal ex-
 801 periments was under 6 vCPU-hours including the gate; no GPUs were used for any measurement
 802 reported in this paper.

803 **Conference-version compatibility.** The conference version’s 191-record QASPER pool, its tim-
 804 ing host, and its numbers remain reproducible from the original artifacts; the journal pools are
 805 supersets drawn under the same filters, and every qualitative conference claim (ordering of methods,
 806 growth of the margin in r , component attribution) reproduces at the larger scale with tighter intervals.
 807 Two conference-version statements are superseded: the QASPER recall intervals, which touched at
 808 their edges at $n = 191$, are separated at $n = 1000$; and the neural-baseline comparison, which was
 809 analytic in the conference version, is measured here.